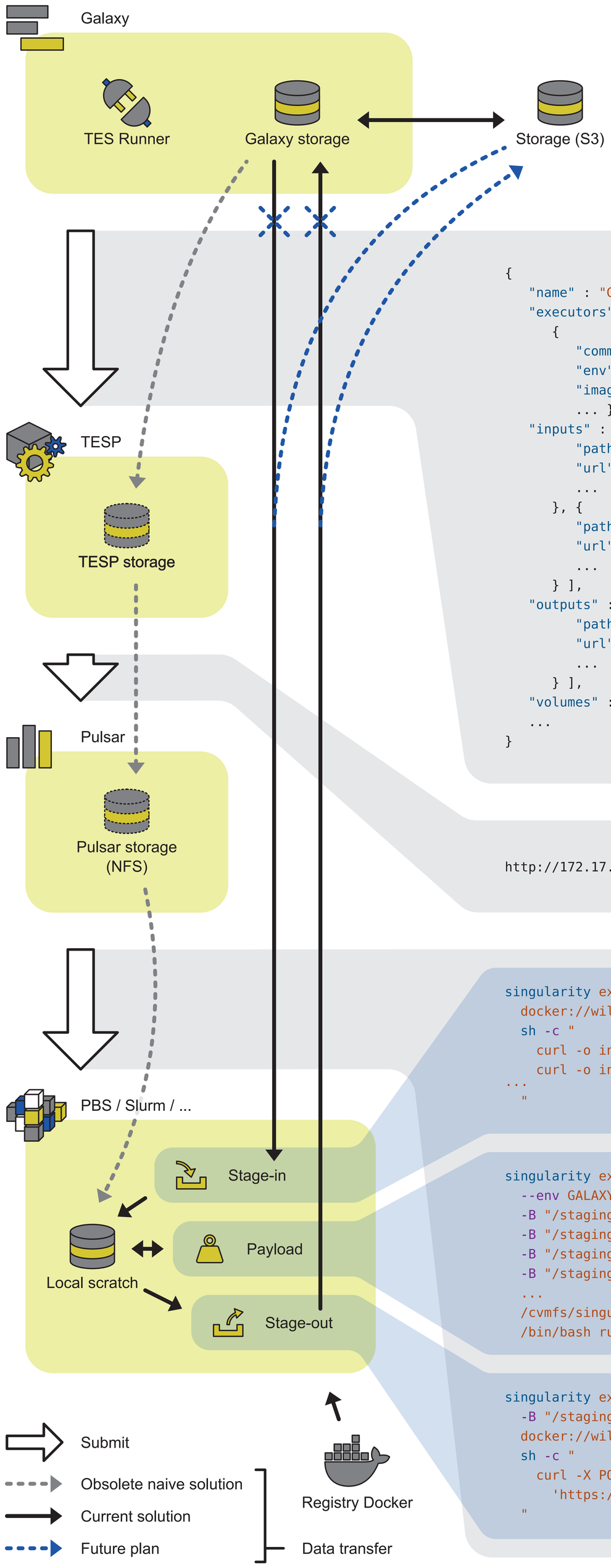
TES job submission and streamlined data transfers



- Extend Pulsar with GA4GH TES interface to accept jobs
- Avoid unnecessary copies of inputs and outputs
- Aim at early production deployment at UMSA Galaxy

```
{
   "name" : "Galaxy Job Execution",
   "executors" : [
      {
         "command" : [ "/bin/bash", "/jobs/11418/tool_script.sh", ..., ],
         "env" : { "GALAXY_MEMORY_MB" : "4000", "GALAXY_SLOTS" : "1", ...  },
         "image" : "/cvmfs/singularity.galaxyproject.org/all/fastqc:0.12.1--hdfd78af_0",
         ... } ],
   "inputs" : [ {
         "path" : "/jobs/11418/tool_script.sh",
         "url" : "https://usegalaxy.cz/api/jobs/9abe73/files?path=/jobs/11418/tool_script.sh",
         ...
      }, {
         "path" : "/jobs/11418/outputs/dataset_e8332a.dat",
         "url" : "https://usegalaxy.cz/api/jobs/9abe73/files?path=/jobs/11418/outputs/dataset_d4e833.dat",
         ...
      } ],
   "outputs" : [ {
         "path" : "/jobs/11418/output.txt",
         "url" : "https://usegalaxy.cz/api/jobs/9abe73/files?path=/jobs/11418/outputs/dataset_fc9a00.dat",
         ...
      } ],
   "volumes" : [ "/jobs/11418" ],
   ...
}
```

```
http://172.17.0.1:8913/jobs/66704e7/submit?command_line=singularity+exec++--pwd+%22/downloads%22+-B%22/staging/…
```

```
singularity exec  --pwd "/downloads" -B "/staging/66704e7/inputs":"/downloads"
  docker://willdockerhub/curl-wget:latest
  sh -c "
    curl -o input_file_0 'https://usegalaxy.cz/api/jobs/9abe73/files?path=/jobs/11418/tool_script.sh' &&
    curl -o input_file_1 'https://usegalaxy.cz/api/jobs/9abe73/files?path=/jobs/11418/outputs/dataset_d4e833.dat' &&
...
  "
```

```
singularity exec  --pwd "/jobs/11418"
  --env GALAXY_SLOTS="1" --env GALAXY_MEMORY_MB="4000" ...
  -B "/staging/66704e7":"/jobs/11418"
  -B "/staging/66704e7/inputs/run_script.sh":"/jobs/11418/run_script.sh"
  -B "/staging/66704e7/inputs/input_file_0":"/jobs/11418/tool_script.sh"
  -B "/staging/66704e7/inputs/input_file_1":"/jobs/11418/outputs/dataset_e8332a.dat"
  ...
  /cvmfs/singularity.galaxyproject.org/all/fastqc:0.12.1--hdfd78af_0
  /bin/bash run_script_0.sh
```

```
singularity exec  --pwd "/downloads"
  -B "/staging/66704e7":"/jobs/11418"
  docker://willdockerhub/curl-wget:latest
  sh -c "
    curl -X POST ...  -F 'file=@/jobs/11418/output.txt'
      'https://usegalaxy.cz/api/jobs/9abe73/files?path=/jobs/11418/outputs/dataset_fc9a00.dat'
  "
```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